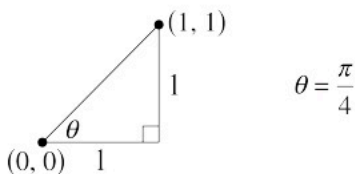


Question 28 (a)

Criteria	Marks
• Provides correct solution	2
• Finds area of the sector	1

Sample answer:

Area = Area sector – Area triangle



$$\begin{aligned}\text{Area} &= \frac{1}{2}r^2\theta - \frac{1}{2} \times 1 \times 1 \\ &= \frac{1}{2}(\sqrt{2})^2 \times \frac{\pi}{4} - \frac{1}{2} \\ &= \frac{\pi}{4} - \frac{1}{2}\end{aligned}$$

Question 28 (b)

Criteria	Marks
• Provides correct solution	2
• Shows that $a = b$	1

Sample answer:

Substitute $x = y = 0$

$$\begin{aligned}0 &= \frac{a}{b} - 1 & \therefore \frac{a}{b} &= 1 \\ \therefore a &= b\end{aligned}$$

Substitute $x = y = 1$

$$\begin{aligned}1 &= \frac{a}{b-1} - 1 \\ 1 &= \frac{a}{a-1} - 1 \\ \frac{a}{a-1} &= 2 \\ a &= 2a - 2 \\ \therefore a &= 2 & \therefore b &= 2\end{aligned}$$

Question 28 (c)

Criteria	Marks
• Provides correct solution	3
• Correctly evaluates integral	2
• Gives the area as a sum of an integral and the area from part (a), or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{Area under curve (hyperbola)} &= \int_0^1 \left(\frac{2}{2-x} - 1 \right) dx \\ &= [-2\ln(2-x) - x]_0^1 \\ &= -2\ln 1 - 1 + 2\ln 2 - 0 \\ &= 2\ln 2 - 1\end{aligned}$$

$$\begin{aligned}\text{Total area} &= 2\ln 2 - 1 + \frac{\pi}{4} - \frac{1}{2} \text{ (units}^2\text{)} \\ &= 2\ln 2 + \frac{\pi}{4} - \frac{3}{2} \text{ (units}^2\text{)}\end{aligned}$$